A prefix code is a type of variable-length code in which no code word is a prefix of any other codeword

A set of binary codewords is said to be a Prefix code iff if for any two distinct codewords C_1 and C_2 neither C_1 is a prefix of C_2 nor C_2 is a Prefix C_1

their importance can be explain in the following points

Instantaneous Decoding

Since no code word is prefix of an another, decoding can be done bit by bit

-> The decoder does not need look-ahead or separators

-> This makes decoding fast and simple

Eliminates Ambiguity

without prefix property, decoding may become ambiguous

Efficient Compression

-> Frequently occuring Symbols are assigned shorter codewords

-> Rare symbols are assigned longer codewords

-> This Reduces the average number of bits per Symbol

-> used in techniques like Huffman coding

Used in practical Compression

Prefix coding in

-> Huffman - coding

-> shannon - Fano coding

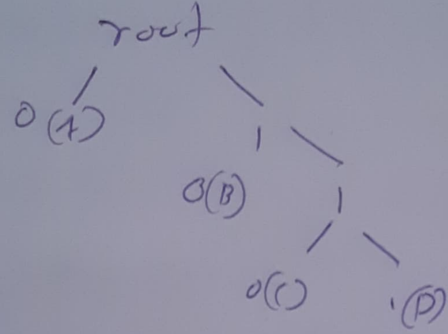
-> Arithmetic coding

they are widely used in ZIP, JPEG, MPEG

Prefix code Representation Using Binary

Left branch - 0
Right branch - 1
Leaf node -> Symbol

Example tree for above code:



Every Symbol is located at a leaf node

Relation with Huffman coding

-> Huffman coding generates an optimal prefix code
-> It minimizes the average code length
-> it is widely used in lossless compression

Advantages of Prefix code

- Unambiguous decoding
- No need for special separator.
- Fast decoding
- Efficient storage
- Foundation of modern techniques

Conclusion

A prefix code is a variable-length coding scheme in which no codeword is a prefix of another. this property ensures instantaneous and unambigous decoding, making prefix coding highly important in lossless data compression. Techniques like Huffman coding use prefix codes to achieve efficient compression by assigning shorter codes to frequent symbols.

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Example of a prefix code

Consider four symbols

A - 0

B - 10

C - 110

D - 111

As we can see

0 is not a prefix of any other codewords

Same goes for 10 and 110

∴ it is a prefix code

Decoding

Consider 01101110

$$\underbrace{0}_{A}\quad \underbrace{110}_{C}\quad \underbrace{111}_{D}\quad \underbrace{10}_{B}$$

Decoded Message ACDB